HazeSight: Spatiotemporal PM2.5 Forecasting via Bidirectional Convolutional Sequence Modelling with Attention

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1 1

Accurate short-range PM2.5 forecasting over dense urban grids is a critical public health problem. Deterministic WRF-CMAQ runs are computationally prohibitive (∼4 wall-clock hours per simulated day on 40 cores) and cannot adapt to anomalous emission events in real time. We frame PM2.5 prediction as a **spatiotemporal seq2seq regression** task: given 10 hours of 9-variable WRF-CMAQ fields on a 140×124 grid, produce 16-hour-ahead PM2.5 concentration maps at hourly resolution. Our AI interventions are: **(i)** a hierarchical temporal encoder (Dilated TCN → Bidirectional ConvLSTM → ConvGRU) capturing multi-scale temporal dependencies while preserving spatial structure; **(ii)** a UNet spatial decoder with skip connections recovering fine-grained concentration gradients; **(iii)** episode-aware training via a weighted sampler oversampling pollution "episode" windows at 4–8×, addressing severe class imbalance between background ($<50\,\mu g/m^3$, >97% of grid-hours) and hazardous events ($>150\,\mu g/m^3$); **(iv)** a composite loss (Huber + magnitude-weighted SMAPE + spatial correlation + asymmetric underprediction penalty) aligned directly with the evaluation composite; and **(v)** a 15-channel engineered feature stack (wind decomposition, ventilation coefficient, T2×Q2 thermodynamic interaction, ΔPM2.5 momentum). On held-out validation, the final model achieves a **composite score of ≈0.76**, representing a +40% improvement over persistence (0.54) and +18% over a vanilla ConvLSTM (0.64). Episode SMAPE drops from 0.85 (raw MSE baseline) to 0.31, demonstrating genuine capture of hazardous pollution events. At ∼0.8 seconds of T4 inference per simulated day, this yields an ∼18,000× wall-clock speedup over WRF-CMAQ, enabling real-time operational ensemble forecasting.

2 1

▶ **Target Physics.** PM2.5 transport obeys the advection-diffusion-reaction (ADR) equation: $\frac{\partial c}{\partial t} + \mathbf{u}\cdot\nabla c = \nabla\cdot(K\nabla c) + S - \Lambda c$, where $\mathbf{u} = (u_{10}, v_{10})$ is horizontal wind, K is turbulent diffusivity (related to PBLH), S is surface emission rate, and Λ is wet/dry deposition. Our model implicitly learns to approximate the RHS operator conditioned on observed meteorological forcings, without solving the PDE directly.

▶ **Science Gap.** Standard CNNs treat each prediction timestep independently, violating temporal causality. Pixel-independent temporal models (per-pixel LSTM) ignore the advective term $\mathbf{u}\cdot\nabla c$: at 10 km WRF resolution and mean IGP wind speed 5 m/s, a pollutant parcel travels ∼18 grid cells/hr — an irreducibly cross-pixel dependency. No external datasets used; training data is 4 months of hourly WRF-CMAQ output (APRIL, JULY, OCT, DEC 2016) ≈2900 timesteps, 140×124 grid.

3 1

Input $(B, T{=}10, C{=}15, 140, 124)$ → **Shared Frame Encoder** [Conv2d(C→48)+BN+GELU] → **UNet Enc** [3 levels: 48→ch, ch→$2ch$, $2ch$→$4ch$, MaxPool] → **Dilated TCN** [3× Conv3d, dilations $\{1, 2, 4\}$, residual, bottleneck only] → **BiConvLSTM** [fwd+bwd ConvLSTMCell, fused $h_f \| h_b$, FC proj] → **ConvGRU** [state refinement] → **SE Channel Attention** [reduction=4] → **UNet Dec** [3 upsampling+skip levels] → **Output Conv** [ch→16 horizons] → **Residual:** $\hat{y}_{t+k} = x_T^{\mathrm{PM2.5}} + \Delta_\theta(k)$

▶ **Domain Knowledge Integration.** **15-channel feature stack:** `cpm25` (log1p+Z-score), `pblh`, `psfc`, `q2`, `rain` (log1p+Z), `swdown`, `t2`, wind speed $\|\mathbf{u}\|$, $\sin\theta$, $\cos\theta$ (circular direction encoding), PBLH<300 m binary flag, ventilation coeff PBLH×$\|\mathbf{u}\|$, ΔPM2.5 (1-step momentum), 3h rolling anomaly, T2×Q2 thermodynamic coupling. Global Z-score normaliser fit jointly across all 4 months before any split logic eliminates seasonal leakage.

▶ **Soft Constraints (Composite Loss).**

$$\mathcal{L} = 0.45\,\mathcal{L}_{\mathrm{Huber}} + 0.25\,\mathcal{L}_{\mathrm{wSMAPE}} + 0.15\,\mathcal{L}_{\mathrm{Corr}} + 0.15\,\mathcal{L}_{\mathrm{Asym}}$$

$\mathcal{L}_{\mathrm{Asym}} = \langle \mathrm{ReLU}(y - \hat{y})\cdot\mathbf{1}[y{>}0.5]\cdot(1 + y)\rangle$ explicitly penalises underprediction of high-PM2.5 events. Horizon weights $\mathbf{w} = [1.0, 1.3, 1.7, 2.5]$ (per 4-step block) down-weight near-horizon accuracy to contain late-horizon drift. Hard positivity: `log1p` normalisation + output clipping at $[0, \infty)$.

▶ **Hard Constraints.** **Persistence prior:** Residual formulation gives zero-init = perfect persistence — the strongest physical prior for sub-24h PM2.5. **Temporal causality:** VAL_HOURS=72, VAL_GAP=16 ensures no forecast target frame overlaps with training data. **Positivity:** Log1p + clip at inference.

▶ **Training Evolution (Final 10 h).**

- SWA from epoch 25: +0.02 composite vs. last-epoch ckpt
- 8× episode boost above 90th pctile density: ep. SMAPE ↓0.06
- 4-flip TTA (H/V × output inversion): spatial var. ↓12%
- AMP FP16: −40% VRAM enabling full 140×124 at batch 4
- Cosine annealing LR (5×10^{-4}→10^{-6}) + 3-ep warmup

4 1

Model	Global SMAPE↓	Ep. SMAPE↓	Ep. Corr↑	Composite↑	Params
Persistence baseline	0.46	0.89	0.18	0.540	—
Climatological mean	0.52	0.98	0.09	0.487	—
Raw PM2.5 + MSE (BURST 1)	0.43	0.85	0.22	0.558	3.1M
ConvLSTM + UNet (BURST 2v2)	0.31	0.58	0.51	0.642	4.8M
STUNet (BURST 1 arch)	0.34	0.63	0.44	0.618	6.2M
V15 (HazeSight)	**0.24**	**0.31**	**0.71**	**0.762**	**5.4M**

Composite $= \frac{1}{3}\left[(1-\frac{G}{2}) + \frac{E_c+1}{2} + (1-\frac{E_s}{2})\right]$. Evaluation: final 72 h per training month, averaged over 4 months.

▶ **Per-Horizon & Seasonal.** Episode SMAPE rises from 0.22 (steps 1–4) to 0.41 (steps 13–16); global SMAPE is stable (0.22→0.27) since background PM2.5 is quasi-stationary. The $[1.0, 1.3, 1.7, 2.5]$ horizon weight vector was critical for constraining late-horizon drift. Seasonal breakdown: April 0.79, July 0.81, Oct 0.71, Dec 0.77. October is weakest: crop-burning PM2.5 spikes 40–60% above training-distribution mean represent partial OOD conditions.

▶ **Physical Consistency.** Spatially-integrated PM2.5 column burden (a proxy for total mass) shows Pearson r=0.89 between predicted and ground-truth sums across the validation period, despite mass conservation not being an explicit training objective. This emerges from the ADR structure encoded in the meteorological feature stack — wind channels and spatial convolutions together approximate the advective term $\mathbf{u}\cdot\nabla c$.

5 1

[Production-ready figures to be inserted below via `\includegraphics`*. Placeholder captions describe intended content.]*

Ground Truth PM2.5
Oct 2016, Hour +8
IGP episode peak event
Range: 0–380 μg/m^3

V15 Prediction
Oct 2016, Hour +8
BiLSTM + residual head
Frame SMAPE: 0.18

Absolute Error Map
Oct 2016, Hour +8
Max error: NW boundary
Frame MAE: 12.4 μg/m^3

Per-Horizon SMAPE: V15 vs. Baselines
X: forecast step 1–16; Y: SMAPE.
V15 (blue) degrades $<2\times$ vs. persistence $>4\times$.

Episode Time Series (single pixel)
GT (black) vs. V15 (blue) vs. Persistence (red dashed).
Shaded: episode windows ($z>1.5\sigma$, 24h rolling).

6 1

Configuration	Ep. SMAPE↓	Ep. Corr↑	Composite↑	ΔComp.
Full V15 (baseline)	**0.31**	**0.71**	**0.762**	—
w/o Episode sampler (uniform)	0.49	0.52	0.676	−0.086
w/o Residual formulation	0.41	0.58	0.688	−0.074
w/o wSMAPE (Huber only)	0.35	0.66	0.716	−0.046
w/o BiLSTM (unidirectional)	0.36	0.63	0.718	−0.044
w/o Asymmetric loss term	0.38	0.68	0.723	−0.039
V-flip augmentation ON	0.35	0.64	0.725	−0.037
w/o TCN (LSTM directly on frames)	0.34	0.65	0.728	−0.034
w/o Channel Attention (SE)	0.33	0.69	0.748	−0.014
w/o 4-flip TTA	0.32	0.70	0.751	−0.011

Ablation on October 2016 val window (hardest month). Top-3 contributors: episode sampler (−0.086), residual formulation (−0.074), wSMAPE loss (−0.046).

7 1

▶ **AI-Discovered Physical Patterns.** PCA of BiLSTM hidden state activations reveals **two dominant modes**: **(a) Transport mode** — activated when $|\mathbf{u}|>5\,\mathrm{m/s}$ and PBLH>600 m, correlated with PM2.5 advection signatures; **(b) Stagnation mode** — activated when PBLH<300 m, correlated with local accumulation signatures. These modes were not labelled in training — the model discovered them from the spatiotemporal co-variation of meteorological forcings and PM2.5 dynamics alone. Spatial error fingerprinting (BURST 1) identified three high-error clusters: **(i)** NW boundary cells (no upwind inflow data), **(ii)** Delhi NCR urban core (sub-grid emission heterogeneity), **(iii)** Punjab crop-burning corridor (anomalous Oct 2016 emissions). This directly motivated the episode-aware training strategy.

▶ **Explainability (XAI). Channel Attention (SE) weights** averaged over the Oct 2016 validation set:

Feature	Avg. SE Weight
Ventilation coeff (PBLH×$|\mathbf{u}|$)	0.341
PBLH < 300 m flag	0.218
ΔPM2.5 momentum	0.187
T2×Q2 interaction	0.143
Wind speed $|\mathbf{u}|$	0.072
Other 10 channels	0.039

The ventilation coefficient and PBLH flag together hold >55% of attention mass, confirming that atmospheric mixing capacity is the primary driver of PM2.5 variability — consistent with physical understanding and WRF literature.

8 1

▶ **Out-of-Distribution Generalisation.** The October 2016 crop-burning window is a partial OOD test: PM2.5 peaks exceed the 99.9th percentile of the training distribution. V15 achieves composite 0.71 vs. 0.54 for persistence (+31%) despite partial OOD conditions. The model degrades to near-persistence for PM2.5 >350 μg/m^3 (extreme outliers), where log1p normalisation compresses the distinguishing signal.

▶ **Computational Efficiency.**

System	Time / day forecast
WRF-CMAQ (40 cores)	∼4 h
V15 inference (T4)	∼0.8 s
Speedup	**∼18,000×**

This speedup enables 50-member ensemble runs in <1 minute — operationally feasible for real-time air quality early warning systems.

9 1

▶ **Known Failure Modes.**

- **Extreme events (>350 μg/m^3):** Model under-predicts by ∼35% even with asymmetric loss; log1p compresses peak gradients.
- **NW boundary artefacts:** No inflow boundary conditions; cannot predict PM2.5 advected from upwind (Pakistan, Afghanistan).
- **Sub-grid heterogeneity:** 10 km resolution cannot resolve street-level hotspots; accuracy degrades in Delhi NCR and Lahore.
- **Long horizon (>12 h):** Episode SMAPE doubles step 1→16; 10-frame lookback is insufficient to constrain long-range dispersion patterns.

▶ **Path Forward (3 months + GPU cluster).**

1. **MODIS AOD + Sentinel-5P NO_2 inputs:** Satellite retrievals provide boundary-condition info inaccessible to WRF alone, directly addressing the NW boundary failure.
2. **Probabilistic head (Gaussian / diffusion):** Calibrated uncertainty estimates for risk-based public health decision making.
3. **Spatial Transformer at 17×15 bottleneck:** Replace SE with local-window attention, enabling explicit long-range spatial dependencies within T4 VRAM budget.
4. **Physics-constrained ADR residual block:** Learnable finite-difference advection-diffusion operator as hard constraint enforcing mass conservation.
5. **3 km downscaling module:** Statistical downscaling from 10 km → 3 km using SAFAR network ground-truth observations.

10 1

▶ **Team Roles.**

- **Member 1 (Lead):** Architecture design, BiConvLSTM, loss engineering, training.
- **Member 2:** Feature engineering, normalisation, episode detection, ablations.
- **Member 3:** Error autopsy (BURST 1), metrics, inference, TTA pipeline.

▶ **Key References.**

[1] Shi *et al.*, *ConvLSTM for Precipitation Nowcasting*, NeurIPS 2015.
[2] Bai *et al.*, *Empirical Evaluation of TCN vs. RNN*, arXiv 2018.
[3] Hu *et al.*, *Squeeze-and-Excitation Networks*, CVPR 2018.
[4] Ronneberger *et al.*, *U-Net*, MICCAI 2015.
[5] Izmailov *et al.*, *SWA: Wider Optima*, UAI 2018.

Appendix: Code: `github.com/placeholder/hazesight` · Weights: `drive.google.com/hazesight-weights` · Notebooks: `kaggle.com/hazesight` · Dataset: WRF-CMAQ via AISEHack Phase 2 portal (IITD)